

SAI RAHUL ADEPU

Python Automation Engineer — Test Automation, Build Automation & CI/CD

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PROFESSIONAL SUMMARY

Python Automation Engineer with 3+ years building test automation frameworks, build/CI automation pipelines, and device automation tooling across embedded, webOS, and avionics platforms. Cut manual DevOps effort by 60% by automating SoC handling, build pipelines, and Jira workflows; boosted automated test coverage by 40–50%; and reduced build handling time from 2 hours to near-zero through Python-driven orchestration. Skilled in Selenium WebDriver, PyTest/unittest, Jenkins CI/CD, Git/Gerrit, Docker, and Linux, with experience designing reusable automation libraries, multi-device/multi-SoC test frameworks, and automated reporting dashboards. Comfortable applying generative AI tools (ChatGPT, GitHub Copilot) to speed up automation script development and debugging.

CORE TECHNICAL SKILLS

- Test & Build Automation: Python Test Automation Frameworks, Selenium WebDriver, PyTest, unittest, Cucumber (BDD), Multi-Device Test Automation, Build Automation, Automated Reporting & Failure-Analysis Dashboards
- CI/CD & DevOps: Jenkins Pipelines, Git, Gerrit, Docker, Linux/Unix Shell Scripting, Jira, DevOps Automation, GitHub Copilot, ChatGPT
- Python Development: Core & Advanced Python, OOP Design, Modular Automation Library Development, Multithreading, Exception Handling, Logging, argparse, Subprocess, Data Parsing (JSON, XML, YAML, CSV)
- Device & Platform Automation: Embedded Systems Automation, webOS, webOS SDK, webOS TAF/TAS, Raspberry Pi Automation, IoT Automation, Bluetooth & Wi-Fi Automation, HLS Media Streaming Validation
- APIs & Protocols: REST API Automation & Integration, HTNG Protocol, Cisco Control Protocol
- Aerospace & Standards: DO-178C, ARINC 661, Requirements Traceability, SVCA Reviews, Avionics Verification & Validation
- Databases: MongoDB, MySQL
- AI/ML for Automation: Machine Learning, Deep Learning, TensorFlow, CNN, OpenCV, NumPy, Pandas, Matplotlib, Scikit-learn
- Other: JavaScript, TypeScript, C, Node.js, Express.js, QML Automation, Agile Scrum

WORK EXPERIENCE

Honeywell — Bangalore

Python Automation Engineer (Software Engineer)

Dec 2025 – May 2026

- Developed Python-based test automation solutions and reusable framework components to validate ARINC 661 Server Core and Widget Library functionality on DU SIM.
- Built automated test execution modules covering log processing, requirements traceability, automated result analysis, and system diagnostics.
- Designed automation workflows to validate widget behavior, runtime messaging, and display configuration, cutting manual verification effort and improving test reliability.
- Automated defect tracking and reporting by integrating Python scripts with Jira and IBM RQM; used Git/Gerrit for automation code source control and peer review.
- Performed root-cause debugging of automation and system failures in Linux-based test environments.

LG Soft India — Bangalore

Python Automation Engineer — Build Automation Systems

Apr 2025 – Dec 2025

- Led design and development of a Python-based Clean Engineering build automation platform with modular, reusable components for build orchestration and workflow management.
- Built Python automation scripts and libraries for SoC configuration, manifest parsing, repository cloning, patch application, and automated build triggering, cutting manual effort by 60%.
- Engineered scalable multi-device, multi-SoC build and deployment automation frameworks, reducing build handling time from 2 hours to near-zero.
- Integrated automation with Jenkins pipelines for real-time status tracking and failure-analysis dashboards; built automated checks with Python and Selenium to improve release stability.
- Documented automation design patterns and mentored engineers on building reusable, maintainable Python automation code.

Python Automation Engineer — webOS Framework

Feb 2025 – Apr 2025

- Designed and enhanced a scalable Python automation framework supporting multi-device and multi-SoC webOS environments.
- Built automated device provisioning, build deployment, and configuration-management modules driven by device type, SoC, and release version.
- Implemented dynamic flashing and onboarding automation, cutting manual setup effort and improving deployment consistency across devices.

Python Automation Engineer — webOS Embedded Systems

Feb 2024 – Jan 2025

- Developed Python automation frameworks for embedded webOS platforms, improving scalability, reliability, and platform coverage; raised automated test coverage by 40–50%.
- Built IoT automation modules enabling automated connectivity validation, device provisioning, and remote interaction across multiple hardware platforms.
- Automated connectivity workflows including Bluetooth pairing, Wi-Fi/hotspot configuration, and screen sharing across multiple devices.
- Built automated media-validation scripts for video playback, resolution management, audio processing, and HLS streaming; developed device health-monitoring and diagnostics automation.
- Automated device communication and reporting by integrating HTNG and Cisco control protocols across iOS/Android companion apps.

Automation & Developer Tooling Engineer — webOS SDK

Jul 2023 – Feb 2024

- Built developer productivity automation for the webOS developer experience via a VS Code Extension and webOS Studio, including smart application import automation.
- Designed a single-click automated setup tool for CLI and library installation, reducing onboarding time by 70%.
- Ensured automation code quality through testing, debugging, and cross-functional collaboration with QA and platform teams.

PROJECTS

ML Pipeline Automation — Iris Recognition (Python, TensorFlow, CNN)

May 2022 – Oct 2022

- Automated an end-to-end biometric iris recognition pipeline in Python using CNNs, from image preprocessing through evaluation.
- Built automated data preprocessing and feature-extraction scripts with OpenCV and NumPy; automated dataset loading and augmentation with Pandas.
- Automated model training and evaluation on the CASIA Iris Dataset using TensorFlow and Scikit-learn, achieving up to 100% classification accuracy.
- Built automated unit, integration, and system test suites in Python to validate real-time recognition and generalization on unseen inputs.

Test Automation — Full-Stack E-Commerce Platform

Jan 2023 – Aug 2023

- Built a full-stack e-commerce platform handling thousands of concurrent users, with RESTful APIs for users, products, carts, and orders.
- Implemented JWT-based authentication and role-based access control; automated cart/order validation and error-handling workflows.
- Integrated MongoDB and MySQL with automated data validation checks for storage and retrieval consistency.

Radisson Hotels Web Application — Frontend Development

Oct 2022 – Jan 2023

- Developed a fully responsive hotel booking UI with modular, reusable components, reducing future development time by 40%.
- Improved accessibility, load performance, and interactivity, increasing user engagement and client satisfaction.

EDUCATION

B.E. – Electrical and Electronics Engineering, Ganapathy Engineering College 2019 – 2022 | GPA: 8.0

MERN Stack Developer (Certification), Vector India Institute 2022 – 2023